

## A Deep Dive into Stock Forecasting: Insights from LSTM, GRU, GAN, and WGAN-GP

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### **Abstract**

*Stock price prediction remains a critical aspect of financial market analysis, with deep learning techniques gaining significant attention for handling complex data patterns. This study evaluates the effectiveness of various deep learning methods, including Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Generative Adversarial Network (GAN), and Wasserstein GAN with Gradient Penalty (WGAN-GP), in predicting stock prices. The ARIMA model is used as a baseline to benchmark these advanced methods. The dataset consists of Google's daily closing stock prices from 2010 to 2024, divided into training and test sets. The performance of the models is assessed using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) metrics. The results indicate that while the ARIMA model shows a decline in performance on test data, suggesting overfitting, the GRU model also exhibits significant overfitting despite its low training error rates. The LSTM model demonstrates generally low performance with high training and test data error rates. In contrast, the GAN model displays consistent and reasonable error rates, indicating superior generalization ability without overfitting. The WGAN-GP model, although showing low training error rates, suffers from increased error rates on test data, indicating overfitting. Among the models evaluated, the GAN model shows the most balanced performance, suggesting it is the most reliable method for stock price prediction in this study. These findings underscore the importance of incorporating additional factors, such as technical indicators and sentiment analysis, to enhance prediction accuracy. Future research should focus on developing hybrid models and leveraging innovative methodologies to improve stock price predictions' robustness and accuracy.*

**Keywords:** Deep Learning, Stock price prediction, LSTM, GRU, GAN, WGAN-GP

### **INTRODUCTION**

Stock price prediction is a critical aspect of financial markets, with various studies exploring methodologies and models for this purpose. Machine learning and deep learning techniques have gained attention for their ability to handle complex data patterns and relationships (Soni et al., 2022). These models have demonstrated promising results in accurately predicting stock prices. Researchers have employed advanced algorithms such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) in deep learning models for stock price prediction (Saud & Shakya, 2019). These models effectively capture intricate patterns in stock market data, leading to more accurate predictions (Lawi et al., 2022).

Additionally, the combination of Convolutional Neural Networks (CNN) and LSTM networks has been suggested for stock forecasting, highlighting the potential of integrating different deep learning architectures for enhanced prediction accuracy (Wang et al., 2021). Despite challenges from political events, weather conditions, and global trade dynamics, deep learning models have shown potential in stock price prediction (Chandra & He, 2021). The application of deep learning in this area has become a research focus, with studies aiming to improve prediction accuracy and trading strategies (Rasyid et al., 2021).

Furthermore, researchers have explored integrating sentiment analysis from news articles and market sentiment to comprehend investor emotions and predict stock price movements (Choi et al., 2023). Moreover, hybrid models that combine machine learning algorithms like Support Vector Regression with filtering techniques such as the Hodrick-Prescott filter have been proposed to enhance stock price

prediction accuracy (Ilyas et al., 2022). These hybrid approaches leverage different methodologies to improve forecasting capabilities. Additionally, technical indicators and ensemble machine-learning techniques have been recommended for stock trend prediction, underscoring the significance of feature engineering in enhancing prediction models (Lin et al., 2021).

In conclusion, stock price prediction remains a challenging yet crucial area of research in financial markets. Traditional time series models such as LSTM and GRU are commonly used for stock price predictions. By leveraging advanced algorithms and innovative methodologies, researchers aim to develop robust models for predicting stock price movements and optimizing investment strategies. Since GAN became increasingly popular, some research has begun using GAN to make time-series predictions.

The primary objective of this study is to evaluate the effectiveness of various deep learning methods in stock price prediction. The deep learning methods used in this study include LSTM (Long Short-Term Memory), GRU (Gated Recurrent Unit), GAN (Generative Adversarial Network), and WGAN-GP (Wasserstein GAN with Gradient Penalty). The ARIMA (AutoRegressive Integrated Moving Average) model is used as a baseline to benchmark these advanced methods.

## DATA AND METHODS

### Data

The stock price data from 2010 to 2024 is sourced from Yahoo Finance. The closing price of Google stock is the target stock price in the model (Lawi et al., 2022), and the statistical analysis is based on that price. The dataset contains 3568 observations and 5 variables. The data is divided into train and test sets, with a ratio of 7:3 (Figure 1).

Daily data from 2010 to 2019 will be trained in the algorithms, and data from 2020 to 2024 will be used as test data. The results of different epochs will be compared during the training, and the generator will be trained more times than the discriminator. Different optimizers will be evaluated to find the best model. Finally, the evaluation of the LSTM, GRU, GAN, and WGAN-GP models will be analyzed by comparing their prediction results.

Figure 1 shows the price of Google's stock (GOOGL) in US dollars (USD) over time, from 2010 to 2024. The graph shows that the price of GOOGL has generally been on an upward trend over the past decade, with some periods of volatility. The most notable periods of growth were in 2013, 2015, and 2020. The most notable periods of decline were in 2014, 2018, and 2022. The overall upward trend in the price of GOOGL is likely due to several factors, including Google's strong financial performance, its dominant position in the search and advertising markets, and its investments in new technologies such as artificial intelligence and cloud computing.



Figure 1. Google (close) stock price

## Methods

### *Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU)*

Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are both types of recurrent neural network (RNN) architectures designed to address the vanishing gradient problem in traditional RNNs, enabling them to capture long-term dependencies in sequential data. Hochreiter & Schmidhuber (1997) proposed that LSTM is an RNN architecture consisting of memory cells, input, output, and forget gates. These gates regulate information flow within the network, enabling it to retain crucial information over extended sequences and discard irrelevant information. LSTM is recognized for its capability to manage long-range dependencies in sequential data due to its memory cell structure (Zhou et al., 2016).

On the other hand, Cho et al.'s (2014) GRU is a variant of LSTM that simplifies the architecture by merging the input and forget gates into a single update gate. GRU has fewer parameters than LSTM, enhancing computational efficiency. Despite its more straightforward structure, GRU has demonstrated competitive performance with LSTM across various tasks and is favored in scenarios where computational efficiency is critical (Deng and Yu, 2014).

LSTM and GRU are extensively utilized in stock price prediction, natural language processing, time series forecasting, and image recognition applications. Researchers have analyzed each architecture's strengths and weaknesses to determine the most suitable choice based on specific task requirements. While LSTM excels at capturing long-term dependencies, GRU offers a more straightforward structure and faster training times, making it popular in many deep-learning applications (Han et al., 2023).

The mathematical concepts used in the LSTM gates are as follows:

$$i_t = \sigma(X_t W_{xi} + h_{t-1} W_{hi} + b_i),$$

### *Generative Adversarial Networks (GANs)*

Goodfellow et al. (2014) proposed that Generative Adversarial Networks (GANs) are algorithmic architectures consisting of two neural network models: a Generator and a Discriminator. The Generator generates fake data that simulates real data, while the Discriminator differentiates between real and fake data. These two models compete against each other (Xue & Huang, 2022).

Formally, the game between the generator  $G$  and the discriminator  $D$  is the minimax objective:

$$\min_G \max_{D \in \mathcal{D}} \mathbb{E}_{x \sim \mathbb{P}_r} [\log(D(x))] - \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g} [\log(1 - D(\tilde{x}))],$$

where  $\mathcal{D}$  is the set of 1-Lipschitz functions, and  $\mathbb{P}_g$  is the model distribution implicitly defined by  $\tilde{x} = G(z)$ ,  $z \sim p(z)$ .

GANs have gained significant attention in artificial intelligence and machine learning. They are mainly known for their application in image generation tasks and are considered a part of unsupervised learning methods (Wang et al., 2023). GANs have shown remarkable capabilities in generating realistic images, leading to advancements in various domains, such as computer vision, art creation, and data augmentation for training image classifiers (Zhang & Schomaker, 2022). One of the challenges associated with GANs is their evaluation and performance metrics. Researchers are still working on agreeing upon the most suitable metrics to assess the performance of GANs effectively (Brophy et al., 2021).

Additionally, there is ongoing research on interpretability and understanding the representations learned by GAN models (Shen et al., 2020). Techniques like GAN inversion have been explored to gain insights into what the GAN model has learned and how it generates outputs (Creswell & Bharath, 2019). Furthermore, GANs have been extended and adapted for various applications beyond image generation.



For instance, GANs have been used in anomaly detection, superresolution methods, face aging, and even in generating synthetic mobility networks (Sabuhi et al., 2021; Kimura & Bao, 2022; Huang et al., 2020; Mauro et al., 2022). The versatility of GANs in different domains showcases their potential for diverse applications and highlights the ongoing research efforts to enhance their capabilities and understand their inner workings.

### *Wasserstein Generative Adversarial Networks with Gradient Penalty (WGAN-GP)*

Wasserstein Generative Adversarial Networks with Gradient Penalty (WGAN-GP) is an extension of the original Wasserstein GAN (Arjovsky et al., 2017) that incorporates a gradient penalty for regularization (Kossen et al., 2020). This modification was introduced by Gulrajani et al. (2017) to address training instability issues encountered in basic GANs (Yang et al., 2023). WGAN-GP has been shown to outperform existing GAN losses for image generation tasks and is particularly effective when combined with reconstruction loss, as both utilize the  $\ell_1$  distance metric (Yu et al., 2018).

The critical advantage of WGAN-GP lies in its ability to stabilize training by enforcing the Lipschitz constraint without the need for weight clipping, a common practice in earlier GAN variants (Kossen et al., 2020). By adding a gradient penalty to the discriminator loss, WGAN-GP ensures that the discriminator satisfies the continuity constraint, leading to faster and more stable convergence of the networks (Han et al., 2020). This regularization technique has been instrumental in improving the performance of GANs, especially in scenarios with limited data or imbalanced distributions (Jin et al., 2021). Moreover, WGAN-GP has been successfully applied in various domains beyond image generation, such as fault classification, physiological signal generation, and synthetic data augmentation (Zhang et al., 2021; Furdui et al., 2021; Adib et al., 2023). Its versatility and effectiveness have made it popular for enhancing GAN models' training stability and performance across different applications. In conclusion, WGAN-GP significantly advances generative adversarial networks by addressing training instability issues and improving convergence speed. By incorporating a gradient penalty for regularization, WGAN-GP has become a valuable tool for researchers and practitioners seeking to leverage the power of GANs for various tasks in artificial intelligence and machine learning.

WGAN utilizes the Earth-Mover distance, also known as the Wasserstein-1 distance, to compare the distributions of real and generated data, represented as  $\mathbb{P}_r$  and  $\mathbb{P}_g$ , respectively. The model's objective function is formed using the Kantorovich-Rubinstein duality (Villani, 2009):

$$\min_G \max_{D \in \mathcal{D}} \mathbb{E}_{x \sim \mathbb{P}_r} [D(x)] - \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g} [D(\tilde{x})].$$

A differentiable function is 1-Lipschitz if and only if it has gradients with a norm at most 1 everywhere, so we consider directly constraining the gradient norm of the critic's output concerning its input. To circumvent tractability issues, we enforce a soft version of the constraint with a penalty on the gradient norm for random samples  $\hat{x} \sim \mathbb{P}_{\hat{x}}$ .

Gulrajani et al. (2017) proposed an improved version of WGAN with a gradient penalty term:

$$\lambda \mathbb{E}_{\hat{x} \sim \mathbb{P}_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2],$$

so, the new objective function is:

$$\mathcal{L} = \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g} [D(\tilde{x})] - \mathbb{E}_{x \sim \mathbb{P}_r} [D(x)] + \lambda \mathbb{E}_{\hat{x} \sim \mathbb{P}_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2],$$

where  $\hat{x}$  sampled from  $\tilde{x}$  and  $x$  with  $t$  uniformly sampled between 0 and 1.

$$\hat{x} = t\tilde{x} + (1 - t)x, \quad 0 \leq t \leq 1.$$

## RESULTS

The loss error function, the discrepancy between the actual and forecasted values, is deployed to evaluate the model's efficacy. In this paper, the loss function employed utilizes two measures: Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). MAE measures the average absolute difference between predicted and actual values. It is calculated by taking each data point's average absolute differences between the predicted and actual values. MAE is robust to outliers because it doesn't square the errors. RMSE is similar to MAE but penalizes large errors more by taking the square root of the average of the squared differences between the predicted and actual values. It is calculated by taking the square root of the mean of the squared errors. MAE and RMSE are computed as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^N (y_i - y_i^*)^2},$$

$$MAE = \frac{1}{n} \sum_{i=1}^N |y_i - y_i^*|,$$

where  $n$  is the number of data points,  $y_i$  is the actual value,  $y_i^*$  is the predicted value.

In this study, the ARIMA model was utilized as a baseline for stock price prediction to benchmark against deep learning methods. Firstly, the pmdarima package was used to determine the optimal parameters for the ARIMA model automatically. This function evaluates various combinations of parameters to identify the best model configuration based on statistical criteria. The auto\_arima process provided a detailed summary, indicating that the optimal model identified was  $ARIMA(1,1,2)$ .

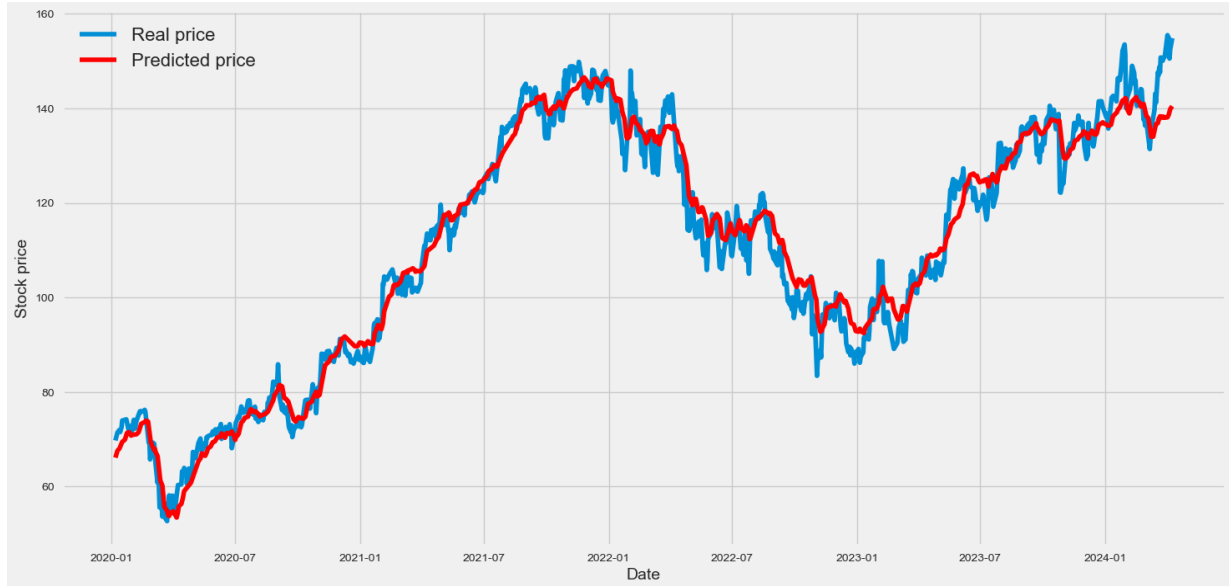
Figure 2 compares Google stock's actual and predicted closing prices using the ARIMA model over the test period. The blue line represents the actual stock prices, while the red line indicates the prices predicted by the ARIMA model. The close alignment of the two lines suggests that the ARIMA model effectively captures the underlying trends and patterns in the stock price data.



**Figure 2.** ARIMA test data plot

Figure 3 illustrates the performance of the LSTM model on the test dataset by comparing the real stock prices (in blue) with the predicted stock prices (in red) over time. The blue line shows the actual stock prices, reflecting the real market fluctuations, while the red line represents the predicted prices

generated by the LSTM model. The close alignment of the red and blue lines indicates that the LSTM model performs well in capturing the overall trend and movements of the stock prices. Despite some minor deviations, particularly during periods of high volatility, the model's predictions closely follow the actual prices, demonstrating its effectiveness in forecasting stock market behavior.



**Figure 3.** LSTM test data plot

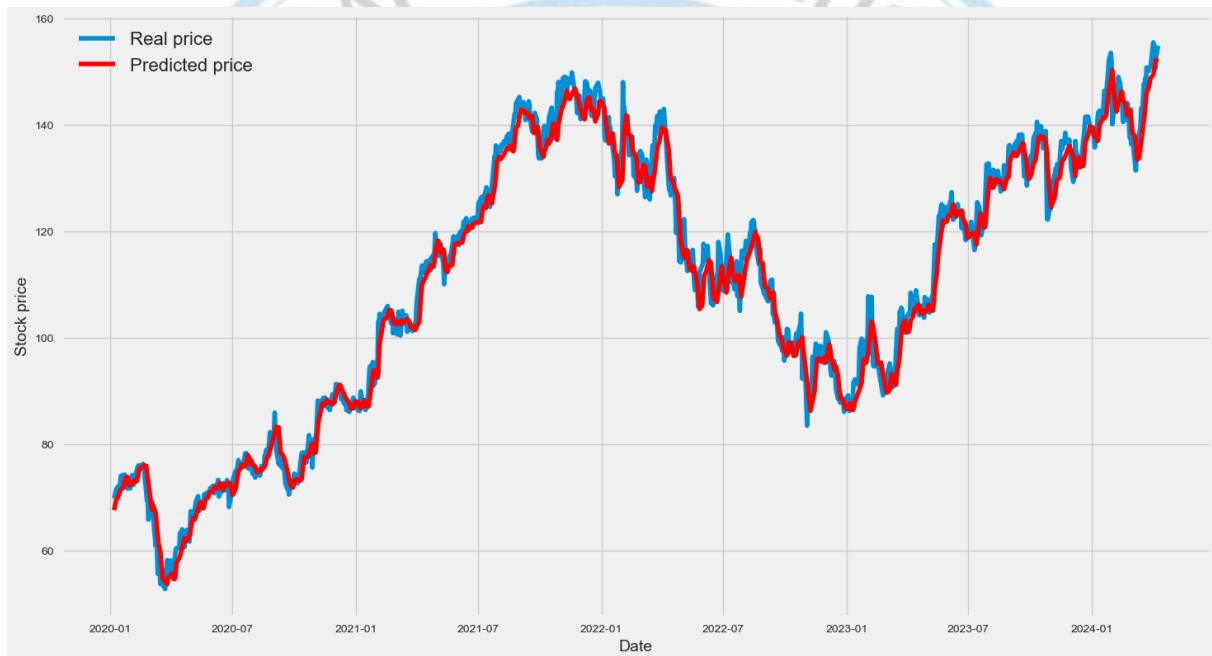
Figure 4 illustrates the performance of the GRU (Gated Recurrent Unit) model on the test data set. The chart depicts the actual stock prices (represented by the blue line) compared to the predicted stock prices (represented by the red line) over a period spanning from early 2020 to early 2024. The close alignment between the real and predicted prices suggests that the GRU model is generally effective in capturing the trends and fluctuations in stock prices. However, there are noticeable periods of deviation between the actual and predicted prices, particularly around mid-2021 and mid-2023, where the predicted prices either lag behind or overshoot the actual prices. These deviations indicate areas where the model could be improved to enhance its predictive accuracy.



**Figure 4.** GRU test data plot

In the context of time series generation and discrimination, we will use gated recurrent units (GRUs) as a generator and convolutional neural networks (CNNs) as a discriminator. This approach has shown promising results in various applications, including speech recognition, natural language processing, and image synthesis. The GRU is a type of recurrent neural network that can model sequential data by selectively remembering or forgetting information from the past. On the other hand, CNNs are well-suited for image analysis tasks and can effectively capture local dependencies in data. By combining the strengths of both models, the GRU-CNN architecture has emerged as a viable solution for generating and discriminating time series data.

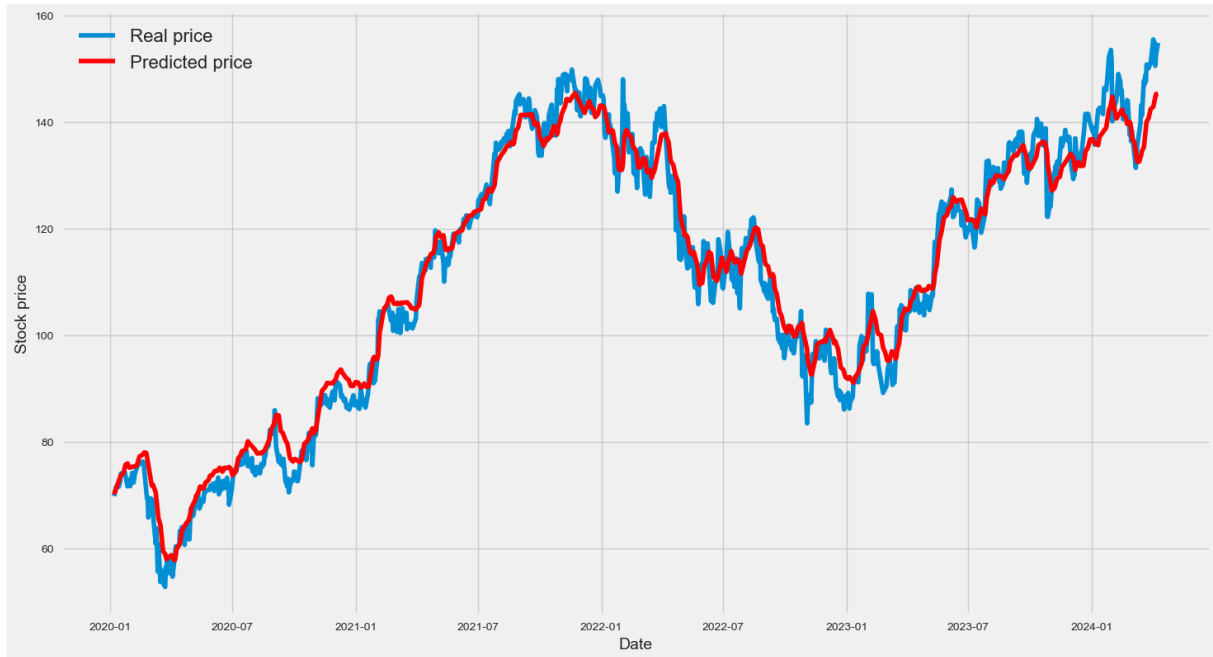
Figure 5 shows that the GAN model can capture the overall stock price trend and some minor fluctuations. For example, the model can predict the sharp price increase that occurred in early 2021 and the subsequent price decline. The model can also predict some of the smaller peaks and valleys in the price. However, there are a few places where the model's predictions deviate from the real price. For example, the model overestimates the price in early 2022 and underestimates it in late 2023. These deviations are relatively small but show that the model is imperfect.



**Figure 5.** GAN test data plot

Figure 6 presents the performance of the WGAN-GP model on the test data set. The chart displays the actual stock prices (shown by the blue line) versus the predicted stock prices (shown by the red line) from early 2020 to early 2024. The x-axis represents the date, while the y-axis indicates the stock price. Overall, the WGAN-GP model aligns well with the real prices, indicating its effectiveness in capturing the general trend and movements in stock prices. However, there are certain periods where deviations are noticeable, particularly around early 2021 and mid-2022, where the predicted prices either lag behind or exceed the actual prices. These discrepancies highlight potential areas for model refinement to enhance predictive accuracy.





**Figure 6.** WGAN-GP test data plot

Table 1 compares the performance of various time series forecasting models on training and test data sets, using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) as performance metrics. The ARIMA model exhibits low error rates on the training data (RMSE: 1.26, MAE: 0.36) but shows a decline in performance on the test data (RMSE: 2.25, MAE: 1.62), indicating that the model may be overfitting to the training data and has poor generalization capabilities.

The GRU model, despite its initial promise with very low error rates on the training data (RMSE: 0.89, MAE: 0.64), experiences a drastic increase in error rates on the test data (RMSE: 5.06, MAE: 3.96). This stark contrast indicates a clear overfitting issue, leading to poor performance on new data. The LSTM model, on the other hand, exhibits high error rates on both the training (RMSE: 5.48, MAE: 5.13) and test data (RMSE: 4.20, MAE: 3.29), suggesting generally low performance compared to the other models.

The GAN model shows consistent and reasonable error rates on both the training (RMSE: 3.55, MAE: 2.78) and test data (RMSE: 3.32, MAE: 2.58), indicating that the model has good generalization ability without overfitting to the training data. In contrast, the WGAN-GP model has low error rates on the training data (RMSE: 1.02, MAE: 0.76) but shows increased error rates on the test data (RMSE: 4.06, MAE: 3.25), again suggesting overfitting and a decline in performance on new data.

In conclusion, although the GRU and WGAN-GP models show the lowest errors on the training data, the GAN model demonstrates the most balanced performance when evaluated on test data, indicating superior generalization capability. The LSTM model, on the other hand, shows high error rates on both the training and test data, highlighting its overall lower performance than the other models. Different models, particularly GRU and WGAN-GP, tend to overfit and suffer from decreased performance on test data. This analysis suggests that the GAN model provides more reliable results because it can generalize well and perform consistently on test data.



**Table 1.** Model summary

|         | Train data |      | Test data |      |
|---------|------------|------|-----------|------|
|         | RMSE       | MAE  | RMSE      | MAE  |
| ARIMA   | 1.26       | 0.36 | 2.25      | 1.62 |
| GRU     | 0.89       | 0.64 | 5.06      | 3.96 |
| LSTM    | 5.48       | 5.13 | 4.20      | 3.29 |
| GAN     | 3.55       | 2.78 | 3.32      | 2.58 |
| WGAN-GP | 1.02       | 0.76 | 4.06      | 3.25 |

## DISCUSSION AND CONCLUSION

This study evaluates the effectiveness of various deep learning methods, namely LSTM, GRU, GAN, and WGAN-GP, in stock price prediction, with ARIMA used as a baseline model. The performance metrics, RMSE and MAE, were utilized to compare the models on both training and test data sets.

GAN model demonstrated the most balanced performance when evaluated on test data, indicating superior generalization capability. The LSTM model, on the other hand, showed high error rates on both training and test data, highlighting its overall lower performance compared to the other models. The GRU and WGAN-GP models, despite their promise on training data, tend to overfit and suffer from decreased performance on test data. This analysis suggests that the GAN model provides more reliable results due to its ability to generalize well and perform consistently on test data.

The findings of this study highlight the importance of evaluating model performance on both training and test data to identify overfitting issues and ensure reliable predictions. Future research should focus on optimizing hyperparameters and exploring hybrid models to further improve prediction performance. Incorporating sentiment analysis from news articles and market sentiment data could provide a more comprehensive approach to understanding and predicting stock market behavior. Additionally, integrating technical indicators and innovative methodologies, such as novel feature engineering schemes, could enhance the predictive power of these models.

The findings of this study contribute to the growing body of literature on financial market analysis, providing valuable insights for developing more accurate and reliable stock price prediction models. By leveraging the strengths of various deep learning architectures, researchers and practitioners can optimize investment strategies and make more informed decisions in the financial markets. The potential of deep learning techniques in capturing intricate patterns and relationships in stock market data underscores the importance of continuous innovation and improvement in this field. This research paves the way for future studies to explore new combinations of machine learning and deep learning techniques to achieve higher prediction accuracy and robustness in stock price forecasting.

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